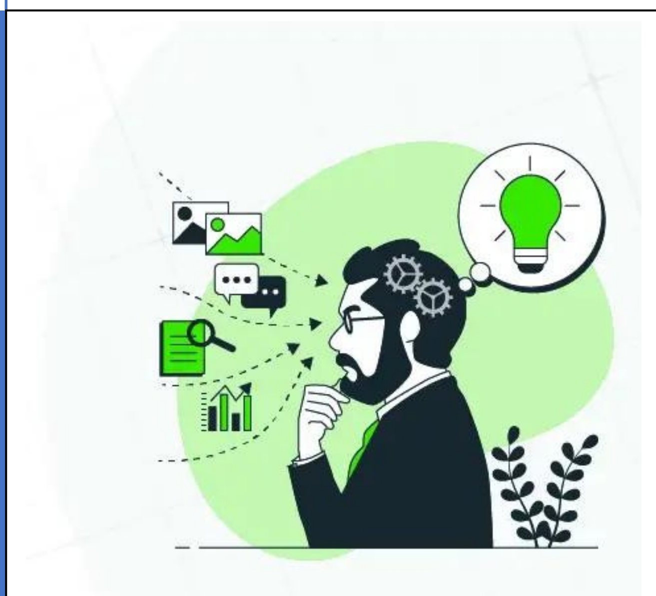
SYSTEM ANALYSIS AND PROJECT MANAGEMENT

In 1 Hour (Revision Guide)



gajendra awasthi

BRANCH 1: SYSTEM FUNDAMENTALS

System Definition

- Interdependent components linked by plan
- Achieves specific objective

System Characteristics

- **Organization**: Structure & order
- **Interaction**: Components communicate
- **Interdependence**: Output = input for another
- **Integration**: Parts coordinated as whole (holism)
- **Central Objective**: Common goal

Information System (IS)

- Collects, processes, stores, distributes info
- Supports decision-making

IS Types

- TPS (Transaction Processing)
- MIS (Management Info)
- DSS (Decision Support)
- EIS (Executive)
- ES (Expert Systems)
- CCS (Communication/Collaboration)
- OAS (Office Automation)

TPS vs DSS

- TPS: Processes transactions, stresses EFFICIENCY
- DSS: Analytical tools, stresses EFFECTIVENESS

BRANCH 2: SYSTEM ANALYST

Roles & Responsibilities

- Define & prioritize user requirements
- Analyze data to identify problems
- Draw specifications for developers
- Evaluate systems
- Agent of change (tech advancements)

Skills Needed

- Communication
- Critical Thinking
- Business Analytics
- Technical Analysis (programming, DB, infra)
- Management

Vs System Designer

- Analyst: Business focus (WHAT)
- Designer: Technical focus (HOW)
- Analyst works with users/sponsors
- Designer creates technical blueprints

BRANCH 3: IS BUILDING BLOCKS

5 Components

- Hardware
- Software
- Data
- Processes
- People

Processes (Detailed)

- Tasks users/staff perform
- Contains business logic/rules
- Transforms input → output
- Modeled via DFDs
- Central to Interaction & Integration

IS Architecture Role

- Plan for component distribution
- Unified framework
- Logic allocation (data/application/presentation)

BRANCH 4: SDLC (Software Development Life Cycle)

7 Phases

1. **Planning**: Preliminary study, risk analysis
2. **Defining**: Requirements, SRS document
3. **Design**: Architecture, DB, UI
4. **Building**: Coding, integration
5. **Testing**: Unit, system, UAT
6. **Deployment**: Production installation
7. **Maintenance**: Fix bugs, enhancements

BRANCH 5: PROJECT MANAGEMENT

Triple Constraint

- **Scope**: Tasks, features
- **Cost**: Budget, resources
- **Time**: Schedule, timeline
- *Adjust one → affects others*

Managing Actions

- Scope: Sign-offs, change orders
- Cost: Estimate tasks, monitor budget
- Time: Gantt charts, resource schedule

System Request Purpose

- Initiates SDLC
- Defines business need & value

Feasibility Analysis

- Determines project viability
- Types:
  - Technical (Can we build?)
  - Economic (Should we build?) - NPV, ROI, TCO
  - Organizational (Will they come?)
  - Legal (Compliance?)

Project Selection Criteria

- Realism
- Capability
- Flexibility
- Ease of Use

- Cost
- Easy Computerization

BRANCH 6: RISK MANAGEMENT

Importance

- Protects objectives
- Maximizes opportunities
- Balances scope/time/cost
- Improves quality
- Informed decisions

7-Step Process

1. **Planning**: Commit approach
2. **Identification**: Threats & opportunities
3. **Assessment**: Likelihood & impact
4. **Strategies**: Avoid/mitigate/transfer/accept (negative); Exploit/share/enhance (positive)
5. **Monitoring**: Scan for triggers
6. **Response**: Execute plan
7. **Evaluation**: Lessons learned

BRANCH 7: REQUIREMENT GATHERING

Definition

- Studying existing system
- Collecting details for improvements

6 Stages

1. Identify Stakeholders
2. Project Definition (goals, scope)
3. Elicitation (interviews, surveys, JAD)
4. Documentation (SRS)
5. Confirmation (approval)
6. Prioritizing (MoSCoW, Kano)

Elicitation Techniques

- Interviews (structured/unstructured)
- Brainstorming
- Focus Groups (6-12 experts)
- JAD (collaborative workshops)
- Shadowing/Observation
- Document Analysis
- Surveys (large groups)
- Prototyping/Wireframing

BRANCH 8: MODELS & DIAGRAMS

DFD (Data Flow Diagram)

- Shows processes & data movement
- Levels: Context (Level 0) → Level 0 → Level 1...
- **Decomposition**: Breaking into detail
- **Balancing**: Consistency across levels

ERD (Entity Relationship Diagram)

- Visual of data created/stored/used
- **Entity**: Person/place/thing (rectangle)
- **Attribute**: Properties of entity
- **Relationship**: Associations (1:1, 1:M, M:N)
- Cardinality defines ratios

Use Case Diagram

- Depicts user interactions
- Represents goals
- Defines functional requirements
- Shows system boundary
- Actors outside, use cases inside

Process Modeling

- Formal representation of operations
- Logical (WHAT) vs Physical (HOW)
- Uses Structured English, Decision Trees/Tables

Data Modeling

- Conceptual → Logical → Physical
- Balanced with DFD via CRUD Matrix

BRANCH 9: ARCHITECTURE

Client-Server

- Balances processing client/server
- **2-Tier**: Client (presentation + app logic), Server (data)
- **3-Tier**: Client (presentation), App Server (logic), DB Server (data)
- **N-Tier**: >3 tiers, web apps

Cloud Architecture

- Hardware + networks + storage + services
- Delivered via internet on demand
- **NIST Actors**: Consumer, Provider, Broker, Auditor, Carrier
- **Service Models**:
    - SaaS (software to users)
    - PaaS (platform for developers)
    - IaaS (infrastructure on demand)
- Relevance: Scalable, handles big data, supports AI

BRANCH 10: USER INTERFACE DESIGN

Why Needed

- Only part users perceive
- Reduces effort, increases speed
- Minimizes errors

4 Principles

1. **Layout**: Consistent screen organization
2. **Aesthetics**: Minimalist, readable fonts, sparing color
3. **Consistency**: Predictable behavior
4. **Usability**: Easy to learn/use

Design Stages

- Wireframing (content/behavior)
- Storyboarding (screen flow)
- Prototyping (mockups)
- Evaluation (heuristic, walkthrough, testing)

BRANCH 11: TESTING

Why Needed

- Verifies system performs as designed
- Ensures error-free
- Meets business needs

Test Types

- **Test Plan**: Document with objectives & cases
- **Unit Test**: Single module
- **Integration/Feature Test**: Modules working together
- **System Test**: Whole system (security, performance)
- **Acceptance Test**: Users (Alpha: fake data, Beta: real data)

BRANCH 12: TRANSITION & MIGRATION

System Transition

- Moving from old to new
- "Move" stage in Lewin's model (Unfreeze → Move → Refreeze)

Conversion Strategies

- **By Style**:
  - Direct (abrupt switch)
  - Parallel (both run)
- **By Location**:
  - Pilot (test location first)
  - Phased (multiple phases)
  - Simultaneous (all at once)
- **By Module**:
  - Whole-system
  - Module-by-module

Post-Implementation Review

- Refreeze stage
- Steps: Handover → Verification → Support → Evaluation → Change Requests

---

BRANCH 13: FORMULAS (QUICK REFERENCE)

Project Selection

- Payback Period = Initial investment / Annual cash inflow
- ARR = Average profit / Average investment
- NPV = Σ (Net Cash Flow/(1+r)^t) - Initial Investment
- Profitability Index = Total benefits / Total cost
- ROI = Profits / Investment cost
- Weighted Score = Σ (Score × Weight)

Scheduling

- PERT Expected Time = (O + 4M + P)/6
- Float = LS - ES or LF - EF

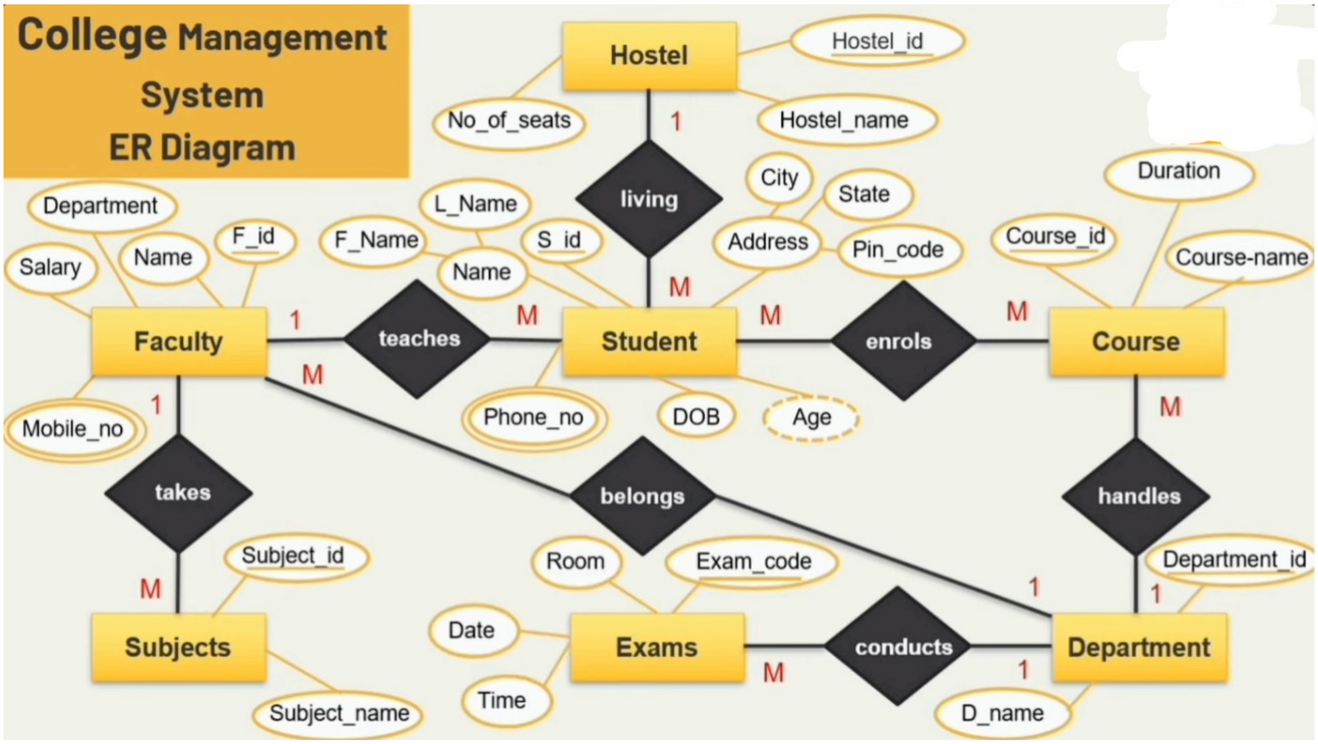
EVM

- CV = EV - AC
- SV = EV - PV
- CPI = EV/AC (≥1 = good)
- SPI = EV/PV (≥1 = good)
- ETC = (BAC - EV)/CPI
- EAC = AC + ETC

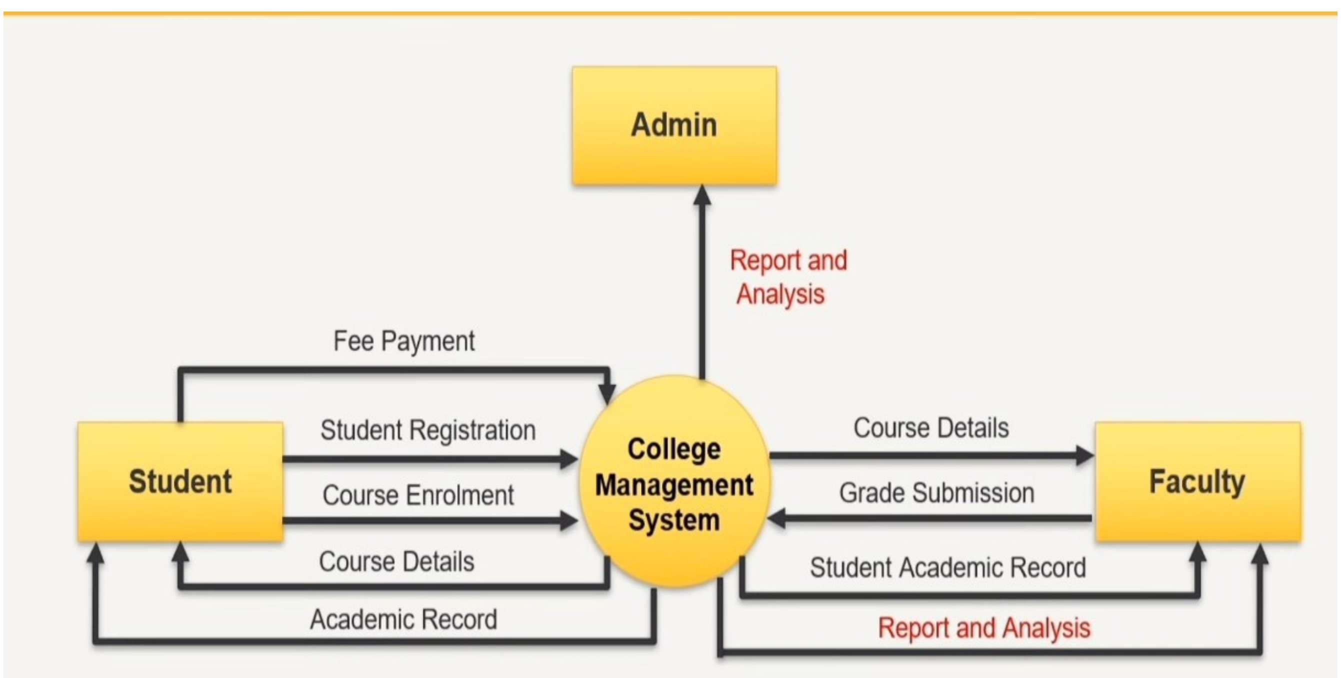
BRANCH 14: KEY DOCUMENTS/TERMS

- **Change Request**: Formal proposal to alter system (bugs, enhancements, integration, environment, strategy)
- **Change Management Plan**: Help users adjust (policies, cost/benefit, motivation, training)
- **Data Dictionary**: Metadata for database (entity name, type, description, attributes)
- **KPI**: Performance indices (CPI, SPI); SMART goals
- **CRUD Matrix**: Maps processes to entities (Create, Read, Update, Delete)

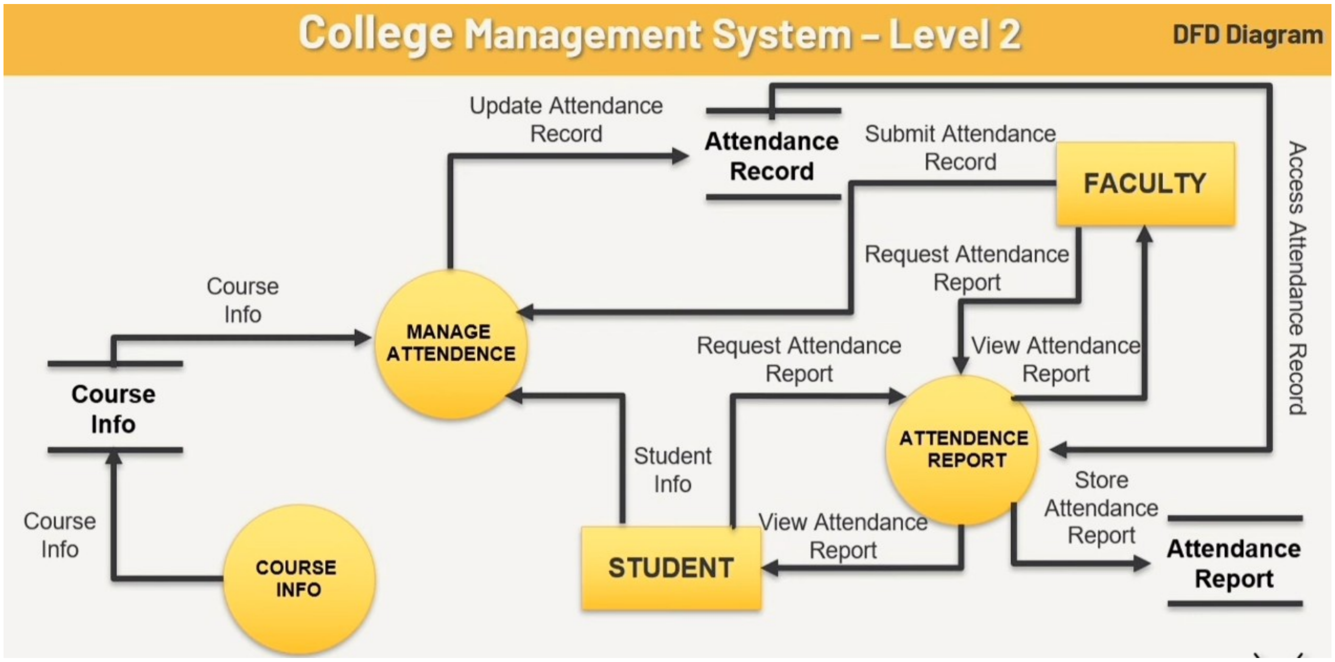
Er Diagram



Context Diagram



DFD



Usecase Diagram